

AI-Powered Hospital Readmission Prevention Program

Project Charter Version 1.0

Healthcare AI Implementation Portfolio Project

Project Overview

Project Type: Healthcare Artificial Intelligence Clinical Decision Support Implementation

Project Sponsor: Chief Medical Officer

Project Lead: Healthcare AI Program Manager

Department: Population Health and Clinical Informatics

Executive Summary

Hospital readmissions are a major healthcare challenge. This project will implement an AI-powered risk prediction solution to identify patients at high risk for 30-day readmission before discharge. The goal is to support early intervention, improve care coordination, and enhance patient outcomes.

Problem Statement

Current identification of high-risk patients depends heavily on manual review and clinical experience. This approach may miss complex patterns in healthcare data. The hospital needs a scalable method to identify patients who may benefit from additional support before discharge.

Project Goal

Implement an AI-based clinical decision support system that predicts patients at high risk for 30-day hospital readmission and supports targeted care interventions.

Project Objectives

- Reduce preventable 30-day readmissions.
- Improve discharge planning and follow-up care.
- Strengthen care coordination.
- Support personalized patient interventions.
- Create a responsible and ethical AI implementation process.

Project Scope

Included:

- AI risk prediction model
- Electronic health record integration
- Clinical workflow design
- Staff training
- AI fairness evaluation
- Performance monitoring

Not Included:

- Replacing clinician judgment
- Autonomous treatment decisions
- Denying patient care based only on AI output

Key Stakeholders

Executive Leadership: Strategy and funding
Physicians: Clinical validation and decision support
Nurses/Care Managers: Patient interventions
Data Scientists: Model development
Health IT Team: Technology integration
Compliance Team: Privacy and regulatory review
Quality Team: Outcome measurement

Business Case

Financial Benefits:

- Potential reduction in readmission-related costs
- Better resource allocation

Clinical Benefits:

- Earlier interventions
- Improved patient experience

Strategic Benefits:

- Builds healthcare AI capability
- Supports value-based care

Major Risks and Mitigation

AI bias: Conduct fairness testing across patient populations.
Clinician resistance: Include stakeholders throughout development.
Alert fatigue: Design appropriate workflows.
Data quality issues: Validate healthcare data sources.
Privacy concerns: Maintain HIPAA compliance.

Success Criteria

The project will be successful when the AI system demonstrates acceptable performance, clinical adoption, improved patient outcomes, reduced preventable readmissions, and equitable performance across patient populations.